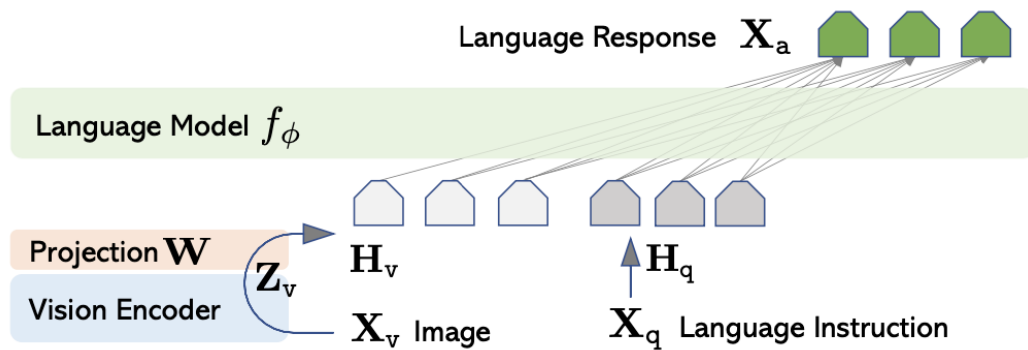


The background of the entire image is a vibrant, multi-colored marbled paper pattern. The colors include deep blues, purples, oranges, and whites, swirling together in a complex, organic design. A white, semi-transparent banner is positioned on the right side of the image, containing text and a decorative element.

LLAVA(09/24/2024)



Large Language and Visual Assistant



$$\mathbf{H}_v = \mathbf{W} \cdot \mathbf{Z}_v, \text{ with } \mathbf{Z}_v = g(\mathbf{X}_v)$$



LLAVA – A LARGE
MULTIMODAL MODEL
(VISUAL+LANGUAGE)



$X_{\text{system-message}} <\text{STOP}> \backslash n$
Human : $X_{\text{instruct}}^1 <\text{STOP}> \backslash n$ Assistant: $X_a^1 <\text{STOP}> \backslash n$
Human : $X_{\text{instruct}}^2 <\text{STOP}> \backslash n$ Assistant: $X_a^2 <\text{STOP}> \backslash n \dots$

Table 2: The input sequence used to train the model. Only two conversation turns are illustrated here; in practice, the number of turns varies based on the instruction-following data. In our current implementation, $X_{\text{system-message}} =$ A chat between a curious human and an artificial intelligence assistant. The assistant gives helpful, detailed, and polite answers to the human's questions. and $<\text{STOP}> = \#\#\#$. The model is trained to predict the assistant answers and where to stop, and thus only green sequence/tokens are used to compute the loss in the auto-regressive model.

TRAINING STRATEGY



- "Describe the image concisely."
- "Provide a brief description of the given image."
- "Offer a succinct explanation of the picture presented."
- "Summarize the visual content of the image."
- "Give a short and clear explanation of the subsequent image."
- "Share a concise interpretation of the image provided."
- "Present a compact description of the photo's key features."
- "Relay a brief, clear account of the picture shown."
- "Render a clear and concise summary of the photo."
- "Write a terse but informative summary of the picture."
- "Create a compact narrative representing the image presented."

Training Strategy Stage 1

$X_{\text{system-message}} <\text{STOP}> \backslash n$
 Human : $X_{\text{instruct}}^1 <\text{STOP}> \backslash n$ Assistant: $X_a^1 <\text{STOP}> \backslash n$
 Human : $X_{\text{instruct}}^2 <\text{STOP}> \backslash n$ Assistant: $X_a^2 <\text{STOP}> \backslash n \dots$

1. Trained on 595k image-text pairs from CC3M (single-turn)
2. Given an image, sample a question from the list of questions, use it as X_{instruct}
3. Caption from the corresponding CC3M image-text pair is the $X_{\text{assistant}}$
4. Everything is frozen except W

GPT4

Vicuna

- "Describe the following image in detail"
- "Provide a detailed description of the given image"
- "Give an elaborate explanation of the image you see"
- "Share a comprehensive rundown of the presented image"
- "Offer a thorough analysis of the image"
- "Explain the various aspects of the image before you"
- "Clarify the contents of the displayed image with great detail"
- "Characterize the image using a well-detailed description"
- "Break down the elements of the image in a detailed manner"
- "Walk through the important details of the image"
- "Portray the image with a rich, descriptive narrative"
- "Narrate the contents of the image with precision"
- "Analyze the image in a comprehensive and detailed manner"
- "Illustrate the image through a descriptive explanation"
- "Examine the image closely and share its details"
- "Write an exhaustive depiction of the given image"

Language Response X_a

Language Model f_ϕ

Projection W

Vision Encoder

Z_v

H_v

X_v Image

H_q

X_q Language Instruction

Training Strategy 1

- Cheap to construct
- But, lacks diversity
- Lack in-depth reasoning
- Solution:
 - Use much stronger **text-only** ChatGPT/GPT4
 - But how to input image information

```
messages = [ {"role": "system", "content": f"""You are an AI visual assistant, and you are seeing a single image. What you see are provided with five sentences, describing the same image you are looking at. Answer all questions as you are seeing the image.""} ]
```

Design a conversation between you and a person asking about this photo. The answers should be in a tone that a visual AI assistant is seeing the image and answering the question. Ask diverse questions and give corresponding answers.

Include questions asking about the visual content of the image, including the **object types, counting the objects, object actions, object locations, relative positions between objects**, etc. Only include questions that have definite answers:

- (1) one can see the content in the image that the question asks about and can answer confidently;
- (2) one can determine confidently from the image that it is not in the image. Do not ask any question that cannot be answered confidently.

Also include complex questions that are relevant to the content in the image, for example, asking about background knowledge of the objects in the image, asking to discuss about events happening in the image, etc. Again, do not ask about uncertain details. Provide detailed answers when answering complex questions. For example, give detailed examples or reasoning steps to make the content more convincing and well-organized. You can include multiple paragraphs if necessary."""

```
for sample in fewshot_samples:
    messages.append({"role": "user", "content": sample['context']})
    messages.append({"role": "assistant", "content": sample['response']})
messages.append({"role": "user", "content": '\n'.join(query)})
```

Could
be
multi-
turns

prompt

Context type 1: Captions

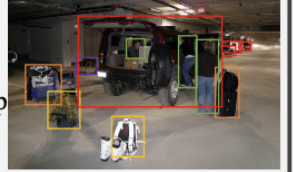
A group of people standing outside of a black vehicle with various luggage.

Luggage surrounds a vehicle in an underground parking area

People try to fit all of their luggage in an SUV.

The sport utility vehicle is parked in the public garage, being packed for a trip

Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], person: [0.63, 0.222, 0.686, 0.516], person: [0.444, 0.233, 0.487, 0.34], backpack: [0.384, 0.696, 0.485, 0.914], backpack: [0.755, 0.413, 0.846, 0.692], suitcase: [0.758, 0.413, 0.845, 0.69], suitcase: [0.1, 0.497, 0.173, 0.579], bicycle: [0.282, 0.363, 0.327, 0.442], car: [0.786, 0.25, 0.848, 0.322], car: [0.783, 0.27, 0.827, 0.335], car: [0.86, 0.254, 0.891, 0.3], car: [0.261, 0.101, 0.787, 0.626]

Response type 1: conversation

Question: What type of vehicle is featured in the image?

Answer: The image features a black sport utility vehicle (SUV).

Question: Where is the vehicle parked?

Answer: The vehicle is parked in an underground parking area, likely in a public garage.

Question: What are the people in the image doing?

Answer: The people are trying to fit all of their luggage into the SUV, likely preparing for a trip.

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip.

In the parking area, various luggage items are scattered around the vehicle. There are two backpacks, one located near the left rear wheel and the other closer to the right side of the vehicle. Additionally, there are two suitcases, one on the right side of the car and another further away near the center of the parking area. A bicycle can also be seen on the left side of the vehicle.

Other cars are parked around the main SUV, with one car positioned behind it and slightly to the left, another behind and slightly to the right, and the third car further behind on the right side.

Response type 3: complex reasoning

Question: What challenges do these people face?

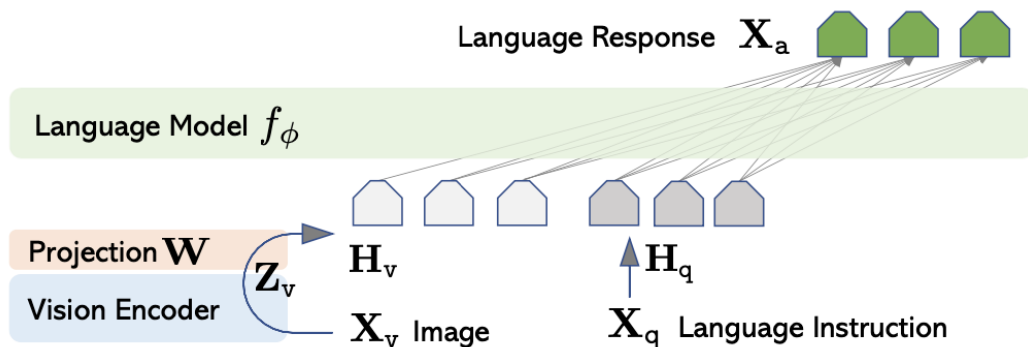
Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings to accommodate. They might have to strategize and arrange the luggage efficiently to ensure that everything fits properly into the vehicle. Additionally, they need to consider the comfort of the passengers and visibility while driving, so the placement of the luggage must not obstruct the driver's view or make the passengers uncomfortable during the trip.

Training Strategy Stage 2

```

Xsystem-message <STOP> \n
Human : Xinstruct1 <STOP> \n Assistant: Xa1 <STOP> \n
Human : Xinstruct2 <STOP> \n Assistant: Xa2 <STOP> \n ...
    
```

1. Fine tune end to end
2. Both W and f_{ψ} (LLM) are updated
3. Multimodal ChatBot – 158k image-language instruction following data
 - Mixture of single and multi-turns
4. Science Q&A dataset
 - Each question is given context in the form of image or text
 - Answers have detailed answers and explanation
 - Multiple choices
 - Single turn



Visual input example, Extreme Ironing:



Source: <https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg>

User	What is unusual about this image?
LLaVA	The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment. [Start a new conversation, and clear the history]
User	What’s happening in the scene?
LLaVA	The image depicts a man standing on top of a yellow SUV in a busy city street. He is holding a portable ladder, seemingly ironing clothes while standing on the vehicle. Around the scene, there are other cars, a traffic light, and a fire hydrant. Additionally, there are two other people in the vicinity, one close to the man on the car and another further away. The man on the ladder appears to be attracting attention due to his unconventional choice of ironing his clothes on top of a moving car . The city street around him is bustling with activity, adding to the unique nature of the scene.
User	What is unusual about this image?
GPT-4 [32]	The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.
User	Can you explain this meme in detail?
BLIP-2	a man is sitting on the back of a yellow cab
User	Can you explain this meme in detail?
OpenFlamingo	The man is drying his clothes on the hood of his car.

Table 4: Example prompt demonstrating LLaVA and GPT-4’s visual input capability. The prompt requires image understanding.

Visual input example, Chicken Nugget Map:

Sometimes I just look at pictures of the earth from space and I marvel at how beautiful it all is.



User	Can you explain this meme in detail?
LLaVA	The meme in the image is a creative and humorous take on food, with a focus on chicken nuggets, as the center of the universe. The meme begins with the phrase "Sometimes I just look at pictures of the Earth from space and I marvel at how beautiful it all is..." The main image features a tray of cooked chicken nuggets, with the chicken pieces positioned to resemble the continents and islands on a world map. The meme playfully suggests that the chicken nuggets represent the Earth, and the various locations depicted in the photo are actually chicken nugget versions of different places. The punchline of the meme appears in smaller text at the bottom, reading "I mean, it’s not the real Earth, but how beautiful it is all is." This lighthearted and entertaining meme combines food and humor, using the image of chicken nuggets to create a funny and imaginative representation of the world.
User	Can you explain this meme?
GPT-4 [32]	This meme is a joke that combines two unrelated things: pictures of the earth from space and chicken nuggets. The text of the meme suggests that the image below is a beautiful picture of the earth from space. However, the image is actually of chicken nuggets arranged to vaguely resemble a map of the world. The humor in this meme comes from the unexpected juxtaposition of the text and the image. The text sets up an expectation of a majestic image of the earth, but the image is actually something mundane and silly.
User	Can you explain this meme in detail?
BLIP-2	sometimes i just look at pictures of the earth from space and marvel how beautiful it is
User	Can you explain this meme in detail?
OpenFlamingo	It’s a picture of a chicken nugget on the International Space Station.

Table 5: Example prompt demonstrating LLaVA and GPT-4’s visual input capability. The prompt requires image understanding.

RESULTS

Evaluation

Quantitative Evaluation. To gain a systematic understanding of the performance of LLaVA, we propose a quantitative metric to measure the model’s instruction-following capability on multimodal data. Inspired by [9], we leverage GPT-4 to measure the quality of generated responses. Specifically, we create triplets consisting of image, ground-truth textual descriptions, and question. The candidate models (*e.g.*, LLaVA) predict the answers based on the question and the image. To provide an *approximate theoretical upper bound*, we create a reference prediction based on the question and the *ground-truth* textual descriptions, using the text-only GPT-4. After obtaining the responses from both models, we feed the question, visual information (in the format of textual descriptions), and the generated responses from both assistants, to the judge (*i.e.*, text-only GPT-4). It evaluates the helpfulness, relevance, accuracy, and level of detail of the responses from the assistants, and gives an overall score on a scale of 1 to 10, where a higher score indicates better overall performance. It is also asked to provide a comprehensive explanation for the evaluation, for us to better understand the

6

	Conversation	Detail description	Complex reasoning	All
Full data	83.1	75.3	96.5	85.1
Detail + Complex	81.5 (-1.6)	73.3 (-2.0)	90.8 (-5.7)	81.9 (-3.2)
Conv + 5% Detail + 10% Complex	81.0 (-2.1)	68.4 (-7.1)	91.5 (-5.0)	80.5 (-4.4)
Conversation	76.5 (-6.6)	59.8 (-16.2)	84.9 (-12.4)	73.8 (-11.3)
No Instruction Tuning	22.0 (-61.1)	24.0 (-51.3)	18.5 (-78.0)	21.5 (-63.6)

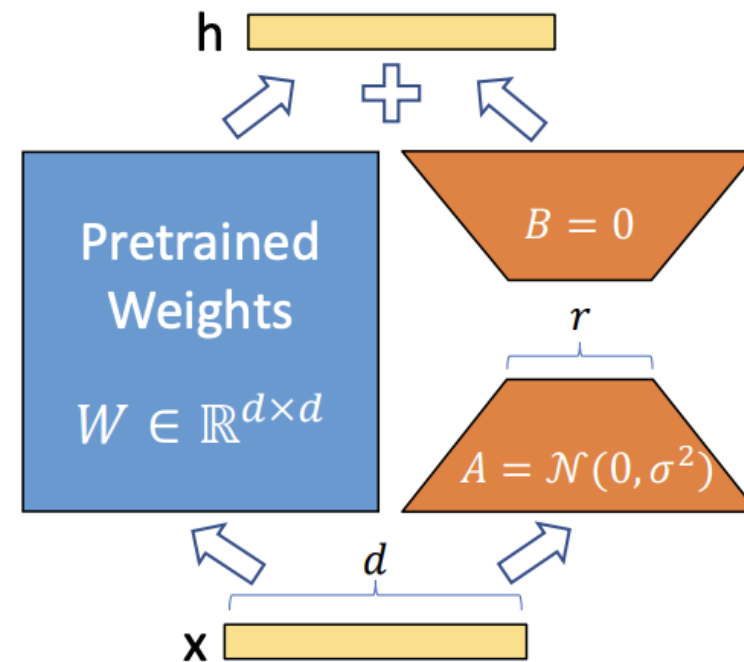
Table 4: Ablation on LLaVA-Bench (COCO) with different training data. We report relative scores *w.r.t.* a text-only GPT-4 model that uses ground truth image captions and bounding boxes as visual input. We prompt GPT-4 with the answers from our model outputs and the answers by GPT-4 (text-only), and let it compare between both responses and give a rating with an explanation.

LORA: LOW RANK ADAPTATION OF LARGE LANGUAGE MODEL



LoRA

- Fine-tuning a Large Language Model (not restricted to LLM) can be challenging:
 - Catastrophic forgetting
 - Each downstream task generates a model the size of a LLM
- LoRA strategy:
 - B and A connected by rank r
 - Storage is $d \times r$ and $r \times k$ versus $d \times k$ -- consider a very large d and k
 - Efficiency – no update on the frozen backbone just the A and B
 - Can add A and B to all layers, some layers, MLP etc



$$W_0 + \Delta W = W_0 + BA, \text{ where } B \in \mathbb{R}^{d \times r}, A \in \mathbb{R}^{r \times k}$$

$$W_0 \in \mathbb{R}^{d \times \bar{k}}$$

ASSIGNMENT

DUE 10/10/2024



- Use LLaVA1.5
 - Awareness: LLaVA-NEXT <https://llava-lmsys.github.io/blog/2024-01-30-llava-next/> Links to an external site. -- no training code
- Use Vicuna-1.5 7B
- Same Human Action Recognition (HAR) dataset
 - Same train/test split
- Add LoRAs, freeze LLaVA and train the LoRAs on:
 - Use LLaMA-3.1-8B to generate questions and answers as taught today for HAR (25%)
 - May not be as good as ChatGPT but it is OK
 - Fine tune LLaVA via LoRAs on the train set (25%)
 - Test on the test set
 - Follow evaluation protocol in Sec. 5.1, Table 4: <https://arxiv.org/pdf/2304.08485> Links to an external site. but use LLaMA-3.1-8B instead of GPT
 - Find LoRA rank that is the smallest but most effective (efficiency) with comparisons to show why (25%)
 - Determine where to add the LoRAs for best performance and explain why with results for comparisons (25%)